COURSE SYLLABUS

Java Programming

25 modules | 100 sessions | live classes, recordings, assignments and placement support

01. Orientation and Java setup (Week 1)

- JDK, IntelliJ IDEA and first program
- How Java runs: JVM, bytecode and memory
- Packages, classpath and project structure
- Course plan, assignments and doubt sessions

02. Language fundamentals (Week 1)

- Primitive types, literals and casting
- Operators, conditionals and switch
- Loops and loop control
- Reading input and formatting output

03. Methods and arrays (Week 2)

- Method signatures, overloading and varargs
- Pass by value and mutable arguments
- Single and multi dimensional arrays
- Lab: marks analyser and matrix operations

04. Strings and the standard library (Week 2)

- String, StringBuilder and immutability
- Common String methods and formatting
- Math, Random and wrapper classes
- Date and time with java.time

05. Object oriented programming (Week 3)

- Classes, constructors and encapsulation
- Inheritance, super and method overriding
- Abstract classes vs interfaces
- Polymorphism with real examples

06. Advanced OOP (Week 3)

- Static members and nested classes
- Records and sealed types
- Enums and constants done right
- Composition over inheritance

07. Exceptions and file IO (Week 4)

- Checked vs unchecked exceptions
- try with resources and custom exceptions
- Reading and writing files
- Working with JSON in Java

08. Collections framework (Week 4)

- List, Set, Map and their implementations
- Comparable, Comparator and sorting
- Iterators and concurrent modification
- Choosing the right collection

09. Generics and streams (Week 5)

- Generic classes, methods and wildcards
- Stream API: map, filter, reduce

- Collectors and grouping
- Lab: refactor loops into streams

10. Functional Java (Week 5)

- Lambdas and functional interfaces
- Method references and composition
- Optional and null safety
- Immutability patterns

11. Multithreading basics (Week 6)

- Threads, runnables and lifecycle
- Race conditions and synchronization
- Executors and thread pools
- CompletableFuture for async work

12. JVM, memory and performance (Week 6)

- Stack, heap and garbage collection
- Profiling basics
- Memory leaks and common pitfalls
- JVM tuning knobs

13. Maven and Gradle (Week 7)

- Project layout and coordinates
- Dependencies and scopes
- Plugins and build lifecycle
- Multi module builds

14. JDBC and MySQL (Week 7)

- Connecting, statements and result sets
- Prepared statements and SQL injection
- Transactions and isolation levels
- Connection pooling with HikariCP

15. Spring Boot foundations (Week 8)

- Starters, auto configuration and profiles
- Controllers and request mapping
- Configuration properties and validation
- Exception handling and error responses

16. Spring Data JPA (Week 8)

- Entities and relationships
- Repositories and query methods
- JPQL and native queries
- Pagination, sorting and projections

17. REST API design (Week 9)

- Resources, verbs and status codes
- DTOs, mapping and versioning
- Pagination, filtering and sorting
- OpenAPI documentation

18. Spring Security (Week 9)

- Security filter chain
- JWT authentication end to end
- Roles, authorities and method security
- Password encoding and CORS

19. Testing Spring applications (Week 10)

- JUnit 5 and assertions
- Mockito for service tests
- MockMvc for controller tests
- Testcontainers with a real database

20. Validation and clean architecture (Week 10)

- Bean validation annotations
- Layered architecture and boundaries
- Mapping with MapStruct
- Logging and audit trails

21. Caching, scheduling and messaging (Week 11)

- Spring cache abstraction
- Scheduled jobs
- Async methods and mail
- Introduction to Kafka or RabbitMQ

22. Docker and deployment (Week 11)

- Dockerising a Spring Boot service
- Compose with MySQL
- Environment profiles for production
- Deploying to a cloud VM

23. Project: banking style REST API (Week 12)

- Domain modelling for accounts and transfers
- Transactions and concurrency safety
- Auth, roles and audit logging
- Tests, documentation and deployment

24. Project: microservice + client (Week 12)

- Splitting the service boundary
- Service to service calls
- React or Thymeleaf client
- Containerised local stack

25. Interview preparation (Week 13)

- Core Java question bank
- OOP and collections deep dive
- Spring interview scenarios
- Coding round and mock interview

Fees, batch dates and mode (online / classroom): confirm the current numbers on the website or with the admissions team before enrolling. Yantrix Technology, Pune.